

1. Diagnosis of CS-related neck pain confirmed by clinical symptoms and imaging findings.³²
2. Age between 18 and 70 years.
3. VAS score ≥ 4 .
4. Symptom duration between 2 weeks and <3 months.

Exclusion Criteria

Participants will be excluded if they meet any of the following criteria:

1. Presence of neurological deficits or serious cervical pathology.
2. Skin lesions or infections in the cervical region.
3. Severe systemic diseases, coagulation disorders, or uncontrolled psychiatric conditions.
4. Recent use of analgesics or receipt of relevant therapies.
5. Pregnancy or lactation.
6. Known allergy to study-related materials.
7. History of cervical surgery or prior acupotomy treatment.

Intervention

Participants will be randomly assigned in a 1:1 ratio to either the acupotomy group or the control group. The intervention period will consist of 3 weeks, followed by a 12-week follow-up period after completion of treatment.

Acupotomy Group: Ultrasound-Guided Acupotomy-Based Treatment Strategy

Participants assigned to the acupotomy group will receive an ultrasound-guided acupotomy-based treatment strategy. The procedure will be performed by experienced physicians who have undergone standardized training in acupotomy techniques. All interventions will be conducted under real-time ultrasound guidance (Wisonic Navi S, C5–1B MHz convex array probe) to facilitate accurate targeting of the intended tissues and enhance procedural safety.

Treatment points will include bilateral Naokong (GB19), bilateral Quyuan (SI13), along with 3–4 additional nodular points identified between these locations based on clinical assessment. All acupoints will be located in accordance with the World Health Organization (WHO) standard acupuncture point system and modern anatomy (Table 1 and Figure 2).

Participants will be placed in the prone position, allowing full exposure of the posterior neck and upper back. After marking the puncture sites, standard sterile preparation and draping will be performed. Under ultrasound guidance, local infiltration anesthesia will be achieved by injecting 1 mL of 0.5% lidocaine at each site. A Type I No. 4 acupotomy needle (1.0 mm $\times$ 40 mm; Figure 3) will then be inserted through the designated entry point.

The needle will be advanced carefully under continuous ultrasound visualization to avoid major vessels and neural structures until it reaches the target site indicated by the arrow in Figure 4. During needle advancement, participants may experience sensations such as soreness or distension. The operator will then perform 2–3 controlled longitudinal and transverse release movements. Adequate release will be judged by a noticeable reduction in resistance at the needle tip and loosening of the surrounding tissue, after which the needle will be withdrawn.

After the procedure, local compression will be applied for approximately 3 minutes to achieve hemostasis, followed by application of a sterile dressing. The intervention will be performed once weekly for a total of 3 sessions.

Table 1 Locations of Acupoints

Acupoints	Locations
Quyuan (SI13)	Located in the scapular region, in the depression at the medial end of the supraspinous fossa of the scapula, approximately at the junction of the medial border of the scapular spine and the medial border of the scapula.
Naokong (GB19)	Located on the head, lateral to the upper border of the external occipital protuberance, approximately 2.25 cun (about 3.7 cm) lateral to the midline, in the depression between the upper border of the origin of the trapezius muscle and the occipital bone.